Waseem Raza, PhD (Electrical and Computer Engineering)

Researcher & Engineering Educator | Intelligent Wireless Systems

Wireless AI | 5G/6G | Network Optimization | Research & Education

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LinkedIn | GitHub | Google Scholar | Coursera Courses | Portfolio

Academic Profile

Electrical and Computer Engineering researcher and educator specializing in AI/ML for wireless networks, resilient learning, and 5G/6G network intelligence. Research spans propagation modeling, generative data augmentation, digital twins, optimization, self-healing, domain adaptation, and reinforcement learning. Academic work includes teaching, laboratory instruction, student mentoring, research publications, proposal development, and curriculum activities. Industrial R&D at AT&T Labs, Futurewei, Impac, and AT&T brings operational data and applied validation to this research program.

Research Interests

Wireless AI and 5G/6G; Robust and Data-Efficient ML; Digital Twins and Propagation Modeling; Network Optimization and Self-Healing; Transfer and Generative Learning; Autonomous Networks and Wearable AI

Doctoral Research Focus

Doctoral research in AI/ML for wireless networks, emphasizing resilient propagation models, data-efficient generative learning, adaptive network optimization, self-healing, and reproducible evaluation.

Education

PhD, Electrical and Computer Engineering *09/2019 - 07/2025*

University of Oklahoma, Norman, Oklahoma, USA

MSc, Telecommunication Engineering *08/2014 - 08/2016*

University of Engineering and Technology, Taxila, Pakistan

BSc, Telecommunication Engineering *11/2010 - 06/2014*

University of Engineering and Technology, Taxila, Pakistan

Relevant Graduate Coursework

LTE Networks and Telecommunications; Emerging 5G Systems; Advanced ML and Computer Vision; Wireless Testbed Design; Numerical Analysis; Information Theory; Digital Signal Processing

Teaching, Research & Professional Experience

Teaching & Instruction

Lecturer, Department of Technology **12/2017 - 09/2019**

University of Lahore, Pakistan *Lahore, Pakistan*

- Taught programming, communication systems, and wireless communication with practical examples.
- Advised senior-design students on projects, academic plans, and early careers.
- Aligned course materials with learning outcomes and accreditation standards.
- Supported accreditation, exams, technical workshops, and student activities.

Teaching areas: Wireless Communication Systems; LTE/5G Networks; Signals and Digital Signal Processing; Machine Learning and AI; Programming and Data Science; Experimental Design and Labs

Student mentoring and research development

- Mentored undergraduate and graduate students in wireless ML experiments, simulation, and technical writing.
- Guided communications thesis and project work, including experiment design and research presentations.
- Advised senior and capstone projects and supported students' academic development.
- Developed practical labs and assignments connecting communication theory with engineering systems.

University Research Appointments

Graduate Research Assistant **09/2019 - 07/2025**

AI4Networks, University of Oklahoma *Tulsa, OK*

- Built 3GPP-compliant Python simulation tools and an AI-enabled 5G research testbed.
- Built GAN, transfer-learning, autoencoder, and reinforcement-learning models for wireless data.
- Led 100+ experiments testing wireless ML under domain shift and limited labeled data.
- Developed self-healing and outage-management methods, improving resilience by 30%.
- Published wireless AI research and mentored junior researchers in experimental design.

Research Associate, ComSens Lab **06/2016 - 12/2017**

COMSATS University Islamabad, Pakistan *Islamabad, Pakistan*

- Modeled wireless communication systems through simulation, analysis, and technical reports.
- Mentored Bachelor's and Master's theses in experiment design and research presentation.
- Reviewed student reports and manuscripts for rigor and publication readiness.
- Supported proposals, literature reviews, lab seminars, and research coordination.

Full-Time Research Scholar, Communication Systems Lab **08/2014 - 02/2016**
University of Engineering and Technology, Taxila, Pakistan *Taxila, Pakistan*

- Researched wireless communication systems while completing an MSc in Telecommunications.
- Supported communications courses, labs, assignments, grading, and examinations.
- Led wireless communication labs and prepared technical manuals and experiments.
- Developed course and student-project materials for practical wireless engineering.

Industry Research & Engineering

Machine Learning Engineer **02/2025 - Present**
AT&T, contractor via Innovatix Inc. *Bedminster, NJ*

- Applied domain-adaptive ML to telecom service assurance and model validation.
- Designed cross-domain evaluation and error-analysis methods for operational ML.
- Worked with engineering and public-safety teams on reliable network assurance.

Data Scientist and AI/ML Engineer **09/2024 - 12/2024**
Impac Exploration Services *Weatherford, OK*

- Built LLM-enabled analytics to interpret geochemical well data for decision support.
- Built Python pipelines in Azure Databricks, Functions, and Synapse Analytics.
- Automated exploration-data ETL, cutting preparation time by 40%.
- Used data-validation and quality controls to improve output reliability by 20%.
- Turned geologists' technical needs into structured datasets and useful analytics.

Advanced-Degrees Intern **06/2024 - 08/2024**
AT&T Labs *Austin, TX*

- Analyzed 500K+ LTE/5G-NR logs with ML, reducing diagnostic time by 20%.
- Improved 5G NR-NTN cell selection by 15% through data-driven validation.
- Presented operational data insights to next-generation wireless engineering teams.

Advanced Wireless Technology Development Intern **09/2022 - 01/2023**
Futurewei Technologies *Schaumburg, IL*

- Improved wireless CSI compression by 25% with transfer learning and autoencoders.
- Ran 50+ experiments on wireless ML robustness under changing data conditions.
- Translated research results into recommendations for future wireless networks.

Advanced-Degrees Intern **06/2022 - 08/2022**
AT&T Labs *Bedminster, NJ*

- Improved 5G standalone user-timing precision by 20% with ML and statistics.
- Analyzed user traces with SQL and presented results to engineering leadership.
- Turned operational traces into indicators for network diagnosis and optimization.

Research Contributions

Publications

Real-World Medication Adherence Monitoring Using Hybrid Neural Networks and Smartwatches *HealthINF | Accepted* 2026

AI-Powered Resilience: A Dual-Approach for Outage Management in Dense Cellular Networks *Computer Communications* 2025 | DOI

On Multi-Parameter Optimization and Proactive Reliability in 5G and Beyond Cellular Networks *Sensors 2025* 2025 | DOI

An AI-Driven Framework for Enhancing Resilience in Propagation Models to Enable Digital Twin *IEEE PIMRC* 2024 | DOI

Refining Wireless Propagation Models using Domain-Informed GANs amid Data Scarcity *IEEE VTC-Fall 2024* 2024 | Link

Data-Driven Intelligent Outage Management for High Shadowing Environments in 5G&B Networks *IEEE SmartNets* 2024

An AI-Enabled Framework to Defend Ingenious MDT-Based Attacks on the Emerging Zero Touch Cellular Networks *IEEE Network, 38(1), 228-237* 2024 | DOI

Positioning Error Impact Compensation through Data-Driven Optimization in User-Centric Networks *IEEE GLOBECOM 2023* 2023 | Link

Toward a Hybrid RF/Optical Lunar Communication System (LunarComm) *IEEE Network* 2022 | Link

Towards Positioning Error Impact Characterization and Minimization in User-Centric RAN *IEEE WCNC* 2022

Unification of RF Energy Harvesting Schemes under Mixed Rayleigh-Rician Fading Channels *AEÜ - International Journal of Electronics and Communications*

2020 | DOI

Research Projects & Platforms

SyntheticNET: 3GPP-Compliant Network Simulation *Research Project | 2020-2024*

Developed a Python-based, 3GPP-aligned simulation environment for realistic mobility, propagation, and cellular-network experimentation when measured data is sparse or costly. Supported controlled data generation and repeatable studies in propagation modeling, network optimization, digital twins, and resilient wireless AI.

Tools & Methods: Python, 3GPP, Wireless Simulation, Mobility Modeling, Propagation Modeling, Synthetic Data, Experiment Design, Digital Twins, Network Analytics

TurboRAN Experimental Platform *Research Infrastructure | 2020-2024*

Built and used a multi-band, multi-tier experimental platform for evaluating AI-enabled radio-access-network methods under controlled network conditions. Supported repeatable measurement collection, testbed experimentation, RAN optimization studies, and validation of emerging wireless-network algorithms.

Tools & Methods: 5G Testbeds, RAN, Network Measurement, Experiment Design, Wireless Prototyping, Multi-Band Systems, Performance Evaluation, RAN Optimization

Propagation Models & Digital Twins *Research Project | 2024*

Developed robust data-driven radio-propagation models for wireless digital twins using domain-informed generative learning and scarce measured data. Combined 3GPP-informed pretraining, GAN-based augmentation, domain adaptation, and statistical validation to improve resilience under distribution shift.

Tools & Methods: Propagation Modeling, Digital Twins, GANs, Domain Adaptation, 3GPP Models, Synthetic Data, Statistical Validation, Wireless AI, Distribution Shift

Wearable Intelligence for Health Monitoring *Research Project | 2024-2026*

Developed deep-learning workflows for recognizing medication-taking and dietary activities from smartwatch accelerometer and gyroscope signals in free-living settings. Evaluated windowing, feature extraction, CNN/LSTM-style sequence models, cross-user generalization, and recall-oriented performance for health-monitoring applications.

Tools & Methods: Wearable AI, Smartwatch IMU, Time Series, Deep Learning, CNN/LSTM, Feature Engineering, Cross-User Evaluation, Activity Recognition, Healthcare AI

AI for Telecom Network & Security Optimization *Certification Project | 2026*

Applied AI concepts to telecommunications network optimization, security analysis, and customer-experience improvement through a structured specialization project. Framed telecom use cases around operational data, AI-assisted analysis, optimization opportunities, and practical service-improvement recommendations.

Tools & Methods: AI for Telecommunications, Network Optimization, Telecom Security, Customer Experience, Python, TensorFlow, Operational Analytics, Service Improvement

Research Methods & Technical Expertise

Machine Learning: Deep learning, transfer learning, domain adaptation, adversarial learning, generative modeling, reinforcement learning, classification, time-series analysis, robustness evaluation, and model validation

Wireless & Networked Systems: LTE/4G, 5G-NR, NTN, mobility, RAN, propagation modeling, outage management, self-healing, network measurements, and service assurance

Research Methods: Experimental design, 3GPP-aligned simulation, statistical analysis, cross-validation, reproducible workflows, distribution-shift evaluation, technical writing, and publication development

Engineering Tools: Python, PyTorch, TensorFlow, scikit-learn, XGBoost, Pandas, NumPy, SQL, cloud analytics, data pipelines, visualization, Git, and Linux

Teaching & Mentoring: Wireless communications, communication systems, programming, machine learning, laboratory instruction, experimental methods, student projects, and research development

Academic Service & Recognition

Academic Service

- Aligned engineering course material with curriculum outcomes and industry requirements.
- Supported accreditation records, examinations, and academic administration.
- Reviewed student reports and research manuscripts, providing publication-focused feedback.
- Participated in research seminars, technical workshops, and student events.
- Contributed to research proposals, literature reviews, and experimental planning.

Honors & Awards

William H. Barkow Scholarship *2024*
University of Oklahoma, Electrical and Computer Engineering

Clyde L. Farrar Endowment Scholarship *2024*
University of Oklahoma

P.H. Robinson Fellowship *2023*
University of Oklahoma

FAS-Charles Forgan Scholarship *2023*
University of Oklahoma

Nettie Vincent Boggs Graduate Endowment *2022*
University of Oklahoma

SHS-Paul D. Massad Scholarship *2022*
University of Oklahoma

Cleo Cross Memorial Scholarship *2021*
University of Oklahoma

Marjorie Haskell Engineering Scholarship *2021*
University of Oklahoma

Fully Funded PhD / Graduate Research Assistantship *2019*
University of Oklahoma

Self-Learning

1. Machine Learning Stanford/Coursera | Completed | Mar 2020
Supervised and unsupervised learning, regularization, model evaluation, and practical predictive modeling.

2. Introduction to Deep Learning & Neural Networks with Keras IBM/Coursera | Completed | Aug 2026
Neural networks, training workflows, optimization, loss functions, and model evaluation with Keras.

3. Introduction to Computer Vision with TensorFlow Google/Coursera | Completed | Aug 2024
Image processing and convolutional neural networks for computer vision using TensorFlow.

4. AI Foundations & Industry Overview for Telecommunication Coursera | Completed | Jun 2026
AI applications in telecom networks, operations analytics, service assurance, and automation.

5. Network and Security Optimization in Telecommunication Coursera | Completed | Jun 2026
Data-driven network optimization, telecom security, reliability analysis, and operational efficiency.

6. 4G Network Fundamentals Mines-Télécom/Coursera | Completed | Aug 2026
LTE radio and core networks, cellular architecture, mobility management, and 4G fundamentals.

7. Generative AI Engineering and Fine-Tuning Transformers IBM/Coursera | Completed | Aug 2026
Transformer architectures, model fine-tuning, embeddings, and generative-AI engineering.

8. 5G Network Fundamentals Mines-Télécom/Coursera | In progress (~60%)
5G NR radio, network architecture, service capabilities, and next-generation cellular fundamentals.